BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 312 · Floating point, binary and Boolean mathematics

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is normalisation's purpose?

Q2. What is first-two-bit rule for normalised two's-complement mantissa?

Q3. State De Morgan: NOT(A AND B).

Q4. Simplify $(A \wedge B) \vee (\neg A \wedge B)$ with working.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

mantissa — Significand representing precision.

exponent — Power-of-two scale.

two's complement — Representation for signed integers.

normalisation — Shift/scaling preserving represented value.

New learning

Understand — Normalising signed binary mantissas shifts bits and compensates exponent to preserve the same value.

Apply — For negative integer conversion, write exactly eight positive bits, invert each, then add one.

Evaluate — For algebraic simplification, show each Boolean law used rather than only the final expression.

Worked example

Normalise $0.0101_2 \times 2^4$.

1. Predict first — Left-shift mantissa once and lower exponent by one.

2. Explain and check — Explain the reason for the result.

We Do · Practise with guidance

Find 8-bit two's complement of −11.

```
+11=00001011
invert=11110100
+1=11110101
```

Predict first

Invert all eight bits then add one, yielding 11110101.

Explain and check

Explain the reason for the result.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Normalise $0.00101_2 \times 2^5$ and express its decimal value. [6]

Question 2

Convert −23 to 8-bit two's complement, showing every step. [6]

Question 3

Simplify $A \vee (\neg A \wedge B)$ using identities; verify for A0,B1. [6]

Exit Ticket · Answer before checking

Normalisation shifts mantissa but never changes exponent.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Standardised mantissa maximising precision.

Q2. The first two bits differ.

Q3. NOT A OR NOT B.

Q4. B∧(A∨¬A)=B.

You Do pages · Independent questions

Q1. Mantissa0.10100 [2]; exponent3 [2]; value .625×8=5 [2].

Q2. +23=00010111 [2]; invert11101000 [2]; add1→11101001 [2].

Q3. (A∨¬A)∧(A∨B) [2]; 1∧(A∨B) [1]; A∨B [1]; substitution gives1 [2].

Exit Ticket · Model answer

False. Exponent compensates for shifting.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: